

Exposé: Multi-Objective Optimisation Methods for Economic Dispatch Energy Models in Calliope

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Abstract

To-Do

Heavily abbreviated, synoptic statement of research need, research question, method, expected result, and benefits.

Keywords: keyword1, keyword2, keyword3, keyword4

1 Introduction

Planning energy systems often requires selecting among different technologies, which affect monetary, emission, reliability, and other types of costs differently. Moreover, these costs frequently conflict with each other, since lower costs are usually associated with higher emissions, and vice versa. Thus, the optimisation result depends not only on the technology configurations, but also on the objective selection.

Calliope is an open-source platform for energy system modelling and optimisation [1]. It mainly serves as a framework to convert the scenario specifications, consisting of technologies, locations, and a grid topology, to a mathematical optimisation problem, extract the solution from a selected solver, and interpret it. Furthermore, this tool allows both optimisation and operation simulation; it allows trying out several scenarios and is simple to configure. The combination of its functionality and reliability made Calliope the tool selected by the Spatial-AI group at Technische Hochschule Deggendorf (THD) for use in EnerPlanET, which supports energy communities in planning future investments like e.g. photovoltaic systems and storage solutions [2, 3].

Despite Calliope only supporting single-objective (SO) optimisation, it can additionally represent several cost classes, such as monetary costs and greenhouse-gas emissions, which are combined into a single scalar objective using a weighted sum.

While semi-multiobjectiveness using the weighted sum method is already provided, it can not always guarantee the ability to find the best solution. Thus, the goal of this Bachelor Thesis Exposé is to observe possible methods for multi-objective (MO) optimisation and, afterwards, suggest possible improvements to the original Calliope framework based on my findings.

2 Background on Single-Objective Optimisation

Calliope formulates an energy system model as a linear programming (LP) [4] or mixed-integer linear programming (MILP) [5] problem. A LP problem consists of a linear objective function and a set of linear equality and inequality constraints. Linear constraints form a convex feasible region in the decision variable space named polyhedron. Since the objective function is a linear combination of decision variables, the feasible region in the objective space also forms a convex polyhedron. Geometrically, the objective can be visualized as a moving hyperplane in the decision variable space, which at the moment of its maximum or minimum has to touch the polyhedron. The touch can happen at a one single point or through a line or another hyperplane, and therefore there can be a single or multiple equally strong solutions to a single optimisation problem.

A MILP problem additionally allows integer decision variables. In the context of energy system planning, integer decision variables usually represent the amount of units of technologies that have to be installed at some locations. By iterating over possible combinations of integer variables the optimizer splits the MILP problem to multiple LP problems, optimises each of them, and picks the best result. Because of this MILP problem is frequently called NP-hard in general case [6]. MILP is exactly the type of problem that Calliope formulates and helps to solve.

From the mathematical perspective, in the standard single-objective case Calliope determines the system configuration and operation that minimise the total monetary cost of installing and operating all technologies:

$$\min_{\mathbf{x} \in X} C^{\text{mon}}(\mathbf{x}) = \sum_{n \in \mathcal{N}} \sum_{k \in \mathcal{K}_n} \left[C_{n,k}^{\text{inv}}(\mathbf{x}) + \sum_{t \in \mathcal{T}} \omega_t C_{n,k,t}^{\text{op}}(\mathbf{x}) \right], \quad (1)$$

where $\mathbf{x}$ is the vector of investment and operational decision variables, and X is the feasible set defined by the technical and energy-balance constraints of the model. The sets $\mathcal{N}$, $\mathcal{K}_n$, and $\mathcal{T}$ denote the model nodes, the technologies available at node n , and the model timesteps, respectively. $C_{n,k}^{\text{inv}}$ represents the annualised investment and fixed operation and maintenance costs of technology k at node n , while $C_{n,k,t}^{\text{op}}$ represents its variable operating cost during timestep t . The factor ω_t is the timestep weight used to account for representative or aggregated timesteps, following the cost aggregation used in Calliope’s mathematical formulation [7].

Calliope uses Pyomo [8] to construct this mathematical optimisation problem based on the technologies, locations, connections, time-series data, costs, and other constraints defined by the model. Pyomo represents the decision variables, constraints, and objective function and passes the resulting LP or MILP problem to a selected external solver. After the optimisation has been completed, Calliope extracts and interprets the solution returned by the solver.

2.1 Multi-Objective Optimisation

The discussion above assumes that the model is optimized with respect to a single objective function. However, energy system planning often involves several conflicting objectives, such as minimizing total system costs and greenhouse-gas emissions. For a model with m objectives, each feasible decision vector $x \in X$ is therefore mapped to an objective vector

$$f(x) = (f_1(x), \dots, f_m(x)). \quad (2)$$

The set of all attainable objective vectors is called the attainable objective set. If X is the feasible region of an LP and all objectives are linear, this set is also convex. In an MILP, however, integer restrictions generally make both the feasible region in the decision variable space and its image in the objective space non-convex.

When the objectives conflict, there is generally no feasible solution that optimises all of them simultaneously. Assuming that all objectives are minimized, a feasible

solution x' dominates another solution x if

$$f_i(x') \leq f_i(x) \quad \forall i \in \{1, \dots, m\}, \quad (3)$$

with a strict inequality for at least one objective. A solution is called Pareto-optimal if it is not dominated by any other feasible solution. The corresponding non-dominated solutions form the Pareto front, which represents the trade-offs between the objectives. Building Pareto front is exactly the main goal in MO optimisation.

In order to build the Pareto front within current Calliope structure using Pyomo, we have to provide a SO MILP problem to the solver. In order to bring the problem from MO to SO, we have to use one of scalarization methods, discussed below.

2.2 Native Implementation of Weighted Sum

Calliope can additionally represent several cost classes, such as monetary costs and greenhouse-gas emissions, by combining them into a single scalar objective using a weighted sum:

$$\min_{\mathbf{x} \in X} \sum_{j \in \mathcal{J}} \lambda_j C_j(\mathbf{x}), \quad (4)$$

where $C_j(\mathbf{x})$ denotes the total system-wide value of cost class j , and λ_j is its user-defined objective weight. Although several cost classes are considered in this formulation, the optimisation problem passed to the solver still contains one scalar objective function. Consequently, one optimisation run returns one optimal system configuration corresponding to the selected combination of objective weights [7].

Although weighted-sum scalarization provides a straightforward way to combine multiple objectives within Calliope's existing optimisation formulation, repeated optimisation with different weights does not necessarily provide a complete or uniform representation of the Pareto front. Geometrically, changing the weights only changes the orientation of the same linear objective hyperplane. A small change in the weights therefore produces only a small change in its slope, which often leaves the same extreme point of the feasible region optimal over a whole range of weight combinations. Since the feasible region of an LP is a polyhedron, optimal solutions are typically found at its vertices, except for special cases in which an entire edge or face is optimal. As a result, many different weight combinations may return the same solution, followed by a sudden switch to another vertex once the objective hyperplane becomes sufficiently tilted. Consequently, the obtained solutions can be unevenly distributed and separated by relatively large gaps, making weighted-sum scalarization poorly suited for a detailed exploration of the entire Pareto front.

A more fundamental limitation follows from the geometry of weighted-sum scalarization itself. Each combination of weights defines a linear objective represented by a hyperplane in the objective space, whose orientation is determined by the relative weights. During minimisation, this hyperplane is shifted towards the attainable objective set until it first touches its boundary. Regardless of its orientation, however, a hyperplane can only support the set from the outside and therefore reach points located on the boundary of its convex hull. If the Pareto front contains a non-convex part, some Pareto-optimal points lie behind the line or hyperplane connecting other attainable points. No choice of weights can make such a point the first point of contact with the weighted-sum hyperplane. These points are therefore unreachable by weighted-sum scalarization, even if an arbitrarily large number of weight combinations is evaluated.

This limitation becomes particularly relevant when Calliope is formulated as a MILP problem. Integer and binary decision variables may produce a discrete and non-convex set of attainable objective vectors, for example when technologies must be installed as indivisible units or selected through binary decisions. Some feasible system configurations may consequently be Pareto-optimal but unreachable through

weighted-sum scalarization. This motivates the investigation of alternative scalarization and MO optimisation methods capable of exploring both supported and unsupported regions of the Pareto front.

3 Scalarization-based Multi-Objective Optimisation

Since the Weighted Sum Scalarization method appears not to be the most effective for MILP in Calliope, let us research alternatives for Scalarization, bringing the MO problem back to SO. This would allow us to keep most of the current Calliope structure, since these methods would just reformulate the LP/acmilp and use the same engine that is already used in Calliope. Since Weighted Sum is already introduced in the previous section, we move on to other methods.

3.1 ε -Constraint Method

The ε -constraint method transforms a MO problem into a scalar optimisation problem by retaining one objective as the primary objective and converting the remaining objectives into parameterised constraints [9]. For a minimisation problem, it can be formulated as

$$\min_{\mathbf{x} \in X} f_k(\mathbf{x}) \quad \text{subject to} \quad f_j(\mathbf{x}) \leq \varepsilon_j, \quad j \neq k, \quad (5)$$

where ε_j defines the maximum acceptable value of objective j . Different Pareto-optimal solutions can be generated by repeatedly solving the problem for different combinations of these bounds. If the original objectives and constraints are linear, their inclusion as constraints preserves the LP or MILP structure of the model.

An important property of the method is its ability to generate both supported and unsupported Pareto-optimal solutions. For every Pareto-optimal solution, there exists a vector of bounds ε for which that solution is optimal. The geometric visualisation, similar to one seen for the Weighted Sum Method in [10], is illustrated in Fig. 1.

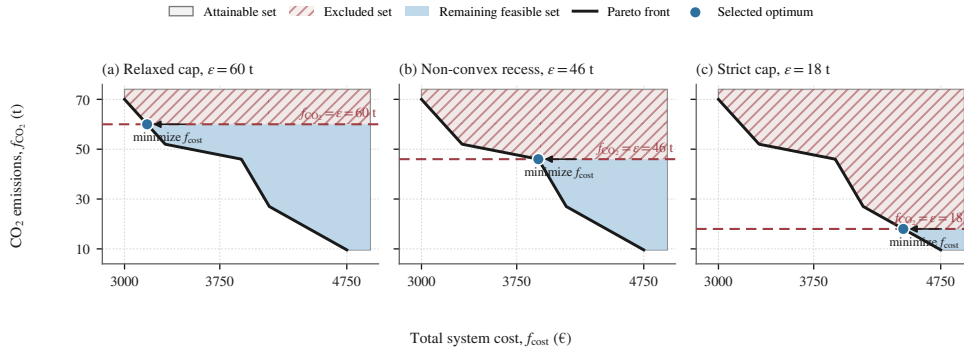


Fig. 1 Geometric interpretation of the ε -constraint method for a non-convex attainable objective set. Tightening the bound on f_2 changes the feasible region and moves the optimal solution along the Pareto front.

However, the conventional ε -constraint method may return weakly Pareto-optimal solutions when alternative optima exist. This is because the constrained objectives are treated only as bounds and are not improved further once these bounds are satisfied [9]. To address this limitation, Mavrotas proposed the augmented ε -constraint method (AUGMECON). For a minimisation problem, the constrained objectives can be written using non-negative slack variables as

$$f_j(\mathbf{x}) + s_j = \varepsilon_j, \quad s_j \geq 0, \quad j \neq k. \quad (6)$$

The normalised slack variables are then included as a lower-priority term in the objective function:

$$\min_{\mathbf{x} \in X, \mathbf{s} \geq 0} \left[f_k(\mathbf{x}) - \rho \sum_{j \neq k} \frac{s_j}{r_j} \right], \quad (7)$$

where r_j is the objective range used for normalisation and ρ is a sufficiently small positive coefficient. Maximising the available slack as a secondary criterion favours improvements in the constrained objectives and thereby prevents the generation of weakly Pareto-optimal solutions.

For generating the Pareto front, the authors also addressed the computational complexity of iterating over all possible ε_j bounds and solving the model repeatedly. If q_j values are tested for each of the $p - 1$ constrained objectives, the maximum number of subproblems is

$$N_{\text{subproblems}} = \prod_{j \neq k} q_j. \quad (8)$$

If the same number q of values is used for every constrained objective, this number becomes q^{p-1} . To reduce the computational effort, the authors included an early-exit mechanism. If a subproblem becomes infeasible after an ε_j bound has been tightened, all subsequent and more restrictive values of this bound will also be infeasible. These values can therefore be skipped. Although this mechanism can significantly reduce the number of subproblems solved in practice, the number of possible combinations may still grow rapidly with the number of objectives [9].

Later, Mavrotas and Florios later proposed an improved version of the method, AUGMECON2 [11], aimed primarily at further reducing the computational effort of the grid search. While the early-exit mechanism of AUGMECON avoids subproblems that are known to become infeasible, AUGMECON2 additionally introduces a bypass mechanism for feasible but redundant grid points. Using the slack obtained from the current solution, the algorithm can identify subsequent ε values that would reproduce the same Pareto-optimal solution and skip the corresponding optimisation runs. This allows a denser exploration of the ε grid with fewer solver calls, while retaining the LP/MILP structure and the ability of AUGMECON to generate efficient solutions.

3.2 Chebyshev / Min–Max Method

The Chebyshev / Min–Max method transforms a MO problem into a scalar optimisation problem by minimising the worst weighted normalised objective value [12]. Let $f_{\text{cost}}(\mathbf{x})$ and $f_{\text{CO}_2}(\mathbf{x})$ denote the original monetary-cost and emission objectives, respectively. Their normalised values are defined as

$$\hat{f}_{\text{cost}}(\mathbf{x}) = \frac{f_{\text{cost}}(\mathbf{x}) - f_{\text{cost}}^{\text{ideal}}}{f_{\text{cost}}^{\text{nad}} - f_{\text{cost}}^{\text{ideal}}}, \quad \hat{f}_{\text{CO}_2}(\mathbf{x}) = \frac{f_{\text{CO}_2}(\mathbf{x}) - f_{\text{CO}_2}^{\text{ideal}}}{f_{\text{CO}_2}^{\text{nad}} - f_{\text{CO}_2}^{\text{ideal}}}. \quad (9)$$

Here, $f_{\text{cost}}^{\text{ideal}}$ and $f_{\text{CO}_2}^{\text{ideal}}$ are the separately attainable minima of the two objectives, while $f_{\text{cost}}^{\text{nad}}$ and $f_{\text{CO}_2}^{\text{nad}}$ are their upper normalization values. Normalisation prevents the different units and numerical ranges of different cost functions from determining their relative importance.

For a cost weight $w \in [0, 1]$, the normalised weighted Chebyshev problem is formulated by introducing an auxiliary continuous variable z :

$$\begin{aligned} \min_{\mathbf{x} \in X, z \geq 0} \quad & z \\ \text{subject to} \quad & w \hat{f}_{\text{cost}}(\mathbf{x}) \leq z, \\ & (1 - w) \hat{f}_{\text{CO}_2}(\mathbf{x}) \leq z. \end{aligned} \quad (10)$$

At the optimal solution, z equals the larger of the two weighted normalised objective values. Therefore, Eq. (10) is equivalent to

$$\min_{\mathbf{x} \in X} \max \left\{ w \hat{f}_{\text{cost}}(\mathbf{x}), (1 - w) \hat{f}_{\text{CO}_2}(\mathbf{x}) \right\}. \quad (11)$$

For a fixed value of z and $0 < w < 1$, the constraints in Eq. (10) can be rewritten as

$$\hat{f}_{\text{cost}}(\mathbf{x}) \leq \frac{z}{w}, \quad \hat{f}_{\text{CO}_2}(\mathbf{x}) \leq \frac{z}{1 - w}. \quad (12)$$

As illustrated in Fig. 2, these bounds define a rectangular sublevel set anchored at the ideal point. Geometrically, increasing z expands this rectangle, while changing w changes its proportions. The optimal value z^* is the smallest value for which the rectangle intersects the attainable objective set.

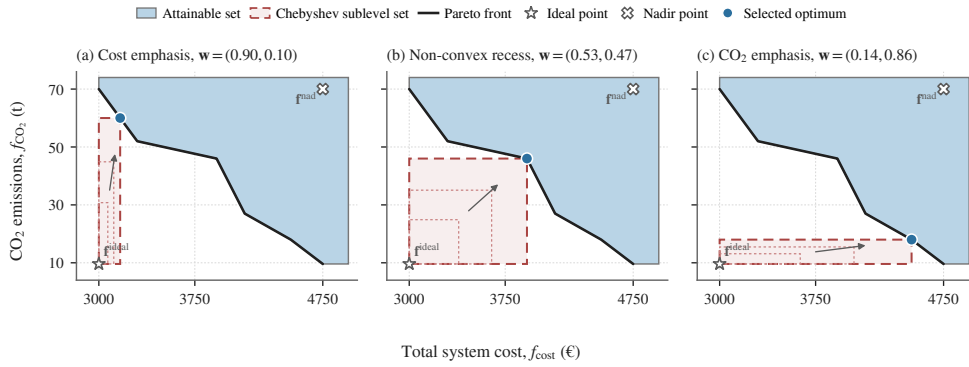


Fig. 2 Geometric interpretation of the normalised weighted Chebyshev method for a non-convex attainable objective set. The nested rectangular sublevel sets illustrate increasing values of the auxiliary variable z , while different weights lead to different trade-off solutions. In the middle panel, the method recovers an unsupported Pareto-optimal solution in the non-convex region.

Repeating the optimisation for different values of w generates different trade-off solutions. Unlike the Weighted Sum Method, the Chebyshev method can recover unsupported Pareto-optimal solutions in non-convex regions of the attainable objective set [12]. If the original objectives and constraints are linear, introducing z and the two bounding constraints preserves the LP or MILP structure.

However, the non-augmented formulation may return a weakly Pareto-optimal solution when several feasible solutions have the same maximum weighted deviation. Geometrically, the method expands a weighted rectangular set from the ideal point until it reaches the feasible region. If the first contact occurs along one of the rectangle's sides rather than at its corner, one objective determines the value of z , while the other objective can still be improved without changing it. As shown in Fig. 3 in panel (a), the standard formulation therefore treats points A and B as equivalent, although B dominates A in the second objective.

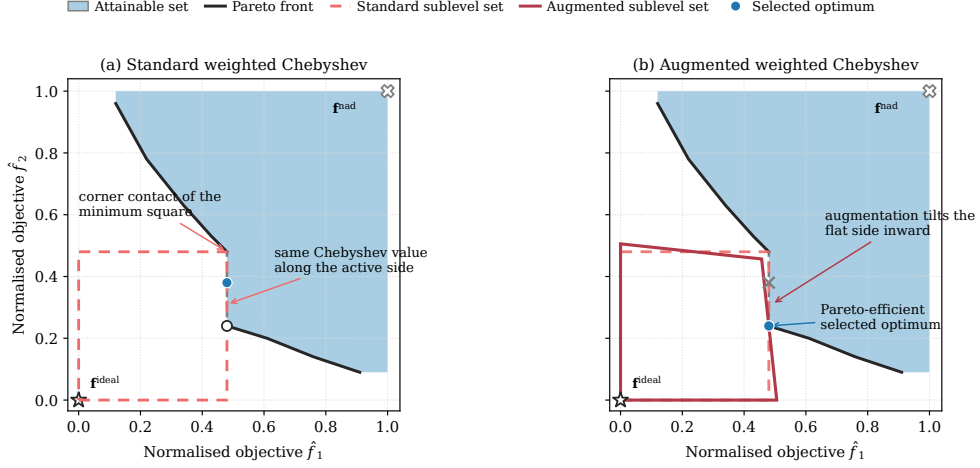


Fig. 3 Geometric effect of augmenting the weighted Chebyshev formulation, illustrated for equal objective weights. In panel (a), the standard rectangular set reaches the attainable set along a vertical side. Points A and B have the same Chebyshev value, although only B is Pareto-efficient. In panel (b), the augmentation term replaces the flat side with a slightly inclined boundary, excludes the weakly Pareto-optimal point A , and selects B .

This ambiguity can be removed by adding a small normalised L_1 -term to the original Chebyshev objective:

$$\begin{aligned} \min_{\mathbf{x} \in X, z \geq 0} \quad & z + \rho \sum_{j \in \mathcal{J}} \hat{f}_j(\mathbf{x}) \\ \text{subject to} \quad & w_j \hat{f}_j(\mathbf{x}) \leq z, \quad j \in \mathcal{J}, \end{aligned} \quad (13)$$

where $\rho > 0$ is the augmentation parameter. The additional term makes the scalar objective sensitive to improvements in objectives that do not currently determine the maximum deviation. In geometric terms, the axis-parallel sides of the Chebyshev sublevel set become slightly inclined. Moving from A to B therefore decreases the augmented objective even though the value of the original maximum term remains unchanged. This guarantees a Pareto-efficient solution [12, 13].

The idea is closely related to AUGMECON: both methods add a small secondary term to distinguish solutions that are equivalent according to the main scalarization criterion. However, their primary mechanisms remain different. AUGMECON uses one selected objective together with bounds on the remaining objectives, whereas augmented Chebyshev minimises the largest weighted deviation from the ideal point. If the objectives and constraints are linear, the augmented formulation preserves the LP or MILP structure. In both cases, the value of ρ should nevertheless be chosen with care: a very small value may become ineffective because of solver tolerances, while an excessively large value may alter the intended Chebyshev trade-off and prevent some nondominated solutions from being generated [9, 13].

3.3 Lexicographic Optimisation

Lexicographic optimisation is not primarily a method for constructing the complete Pareto front. Instead, it is a sequential procedure for selecting one solution according to an explicit order of objective priorities. After a higher-priority objective has been optimised, its value is fixed or relaxed by a specified tolerance before the next objective is considered.

Within the scope of this thesis, lexicographic optimisation may be used in the following cases:

1. **Tie-breaking between alternative optima.** A scalarization method may return several solutions with the same value of its main criterion, including solutions that are only weakly Pareto-optimal. The main criterion can then be fixed at its optimal value while the remaining objectives are optimised sequentially. In this role, lexicographic optimisation provides an explicit alternative to the augmentation terms used in AUGMECON and augmented Chebyshev scalarization.
2. **Sequential optimisation with predefined tolerances.** After optimising a higher-priority objective, a limited deterioration of its result may be permitted before the next objective is considered. For example, monetary cost may first be minimised, after which emissions are optimised while allowing the cost to increase by no more than 5%. The same procedure can be repeated for further objectives, with a separate tolerance defined at each stage.
3. **Optimisation under a predefined objective hierarchy.** Lexicographic optimisation is also suitable when the priorities are established in advance, for example reliability first, emissions second, and monetary cost third. In this case, improvements in a lower-priority objective cannot compensate for an unacceptable deterioration of a higher-priority one.

The method preserves the LP or MILP structure of the original model and requires one sequential optimisation run for each objective. However, a fixed priority order normally produces only one solution and therefore cannot replace augmentation when a broader exploration of the Pareto front is required.

4 Evolutionary Multi-Objective Optimisation

In contrast to the scalarization methods discussed above, Pareto-based evolutionary multi-objective optimisation methods do not transform all objectives into one scalar objective function. Instead, they process the objective functions separately and search for several trade-off solutions within one optimisation process [14]. These methods are usually population-based, meaning that instead of considering only one solution $\mathbf{x}$, they operate on a population P containing N candidate solutions:

$$P = \mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(N)}. \quad (14)$$

Each candidate $\mathbf{x}^{(i)}$ represents one possible combination of decision variables and is evaluated directly according to all m objective functions of the original MO problem:

$$f(\mathbf{x}^{(i)}) = \left(f_1(\mathbf{x}^{(i)}), \dots, f_m(\mathbf{x}^{(i)}) \right). \quad (15)$$

Therefore, instead of assigning every candidate one scalar objective value, the algorithm compares their objective vectors in the m -dimensional objective space.

During the optimisation process, the population is iteratively modified using operations inspired by natural evolution, such as selection, crossover, and mutation. Selection favours candidates that provide better objective values, while crossover and mutation generate new candidate solutions from the existing population. Since no single candidate is generally optimal with respect to all conflicting objectives, the solutions can be compared using the Pareto dominance relation defined in Eq. (3). Non-dominated solutions are preferred over dominated ones and are gradually moved towards the Pareto front over several generations.

In addition to convergence towards the Pareto front, evolutionary methods have to preserve diversity among the generated solutions. Without an additional diversity mechanism, the population may concentrate in only one small part of the objective space and provide an incomplete representation of the available trade-offs. Therefore, multi-objective evolutionary algorithms usually combine a dominance-based selection criterion with a mechanism that distributes the candidates across different regions of the Pareto front [14].

As a result, one execution of a multi-objective evolutionary algorithm can produce a set of different non-dominated solutions instead of one solution corresponding

to a predefined scalarization parameter [15]. Pareto-based evolutionary methods are also not restricted to supported solutions and can therefore approximate non-convex or disconnected parts of the Pareto front. However, the resulting front remains an approximation: the algorithm does not generally guarantee that every returned solution is globally Pareto-optimal or that all relevant regions of the true Pareto front have been identified.

4.1 NSGA-II

One of the most established population-based approaches to MO optimisation is the non-dominated sorting genetic algorithm (NSGA). The original algorithm was introduced by Srinivas and Deb in 1994 [16], while its improved version, Non-dominated sorting genetic algorithm II (NSGA-II), was proposed by Deb et al. in 2002 [17]. The main task of NSGA-II is to decide which candidate solutions should survive from one generation to the next while pursuing two goals at the same time: moving the population towards the Pareto front and keeping the solutions distributed across different trade-off regions.

To determine which candidates are preferable, NSGA-II first compares them using the Pareto dominance relation defined in Eq. (3). All candidates that are not dominated by any other solution in the current population form the first non-dominated front. These candidates are then temporarily excluded, and the same comparison is repeated for the remaining population to construct the second front. The procedure continues until every candidate has been assigned to a front. In this way, each solution receives a Pareto rank: candidates in the first front receive the best rank, candidates in the second front the next one, and so on. During selection, solutions with a better Pareto rank are therefore preferred over those belonging to later fronts.

This sorting step is one of the main improvements of NSGA-II over the original NSGA. Deb et al. reduced its computational complexity from $\mathcal{O}(mN^3)$ to $\mathcal{O}(mN^2)$, where m is the number of objectives and N is the population size [17]. However, Pareto rank alone is still insufficient for selecting between candidates from the same front, since none of these solutions dominates another. If selection were based only on the rank, the population could gradually concentrate in a small part of the objective space and lose important trade-off solutions.

To address this problem, NSGA-II additionally uses the *crowding distance* as a measure of how densely populated the neighbourhood of each candidate is. For every objective, the solutions within one front are sorted according to their objective values. For an internal solution $\mathbf{x}^{(i)}$, its crowding distance can be expressed as

$$d_i = \sum_{j=1}^m \frac{f_j(\mathbf{x}^{(i+1)}) - f_j(\mathbf{x}^{(i-1)})}{f_j^{\max} - f_j^{\min}}, \quad (16)$$

where the neighbouring solutions are defined after sorting by objective j . A larger crowding distance indicates that the candidate lies in a less densely populated region of the objective space and is therefore preferred if two candidates have the same Pareto rank. Boundary solutions are assigned an infinite crowding distance to preserve the extreme points of the current front [17].

To generate the next generation, NSGA-II first creates an offspring population using selection, crossover, and mutation. The parent and offspring populations are then combined, sorted again into non-dominated fronts, and the best N candidates are retained according to their Pareto rank and crowding distance. This elitist selection prevents good solutions found in previous generations from being lost. By repeating this procedure, the population is expected to move towards the true Pareto front while remaining distributed across different trade-off regions. In contrast to the scalarization methods discussed previously, a single NSGA-II run therefore aims to approximate several parts of the Pareto front simultaneously rather than solving one scalar optimisation problem for each selected parameter value.

4.2 Applicability to Calliope

An evolutionary MO algorithm such as NSGA-II can be combined with Calliope in two conceptually different ways. The main difference is what the evolutionary algorithm treats as its decision variables.

The first approach is to optimise the parameters of a scalarization rather than the energy system itself. For example, an individual in NSGA-II may represent the objective weights of a weighted-sum or Chebyshev formulation. Calliope then solves the corresponding scalarized LP/MILP problem and returns the individual objective values to the evolutionary algorithm. The evolutionary search can therefore guide which scalarized problems are solved instead of evaluating a predefined parameter grid. However, the set of solutions that can be obtained still depends on the underlying scalarization formulation. For example, evolutionary selection of weighted-sum parameters does not make unsupported Pareto-optimal solutions reachable.

The second approach is to apply the evolutionary search directly to energy-system decision variables. An individual may, for example, represent installed generation and storage capacities or other investment decisions. For each candidate configuration, Calliope can retain its role of optimising system operation and return the resulting costs, emissions, and other objective values. This creates a hybrid architecture in which the evolutionary algorithm explores system designs while the operational problem remains solver-based. In principle, operational variables could also be included directly in the evolutionary search, but for detailed energy-system models this would considerably increase the number of decision variables and make the representation of technical constraints more difficult.

Both approaches therefore allow evolutionary MO algorithms to be combined with Calliope, but they operate on different search spaces. The first searches the parameter space of a solver-based scalarization, whereas the second searches the space of energy-system configurations directly. In both cases, many candidate solutions may have to be evaluated through repeated Calliope optimisations. Their practical performance, however, depends on the specific formulation, search strategy, and energy-system model and cannot be inferred from this computational structure alone.

5 State of the Art & Related Work

Energy-system models are commonly formulated as LP or MILP problems, allowing specialised solvers to determine an optimal system configuration for a given scalar objective. Extending such models to several conflicting objectives introduces a different challenge: instead of obtaining a single optimum, the optimisation process has to explore a set of trade-off solutions. For the scope of this work, three approaches in the literature are particularly relevant. The first preserves the LP/MILP structure of the original model and generates multi-objective solutions by solving a predefined sequence of scalarized optimisation problems. The second retains the same solver-based scalarization framework but introduces adaptive or evolutionary methods to guide the selection of scalarization parameters, such as objective weights. The third moves the multi-objective search to the energy-system decision variables themselves, using evolutionary algorithms to explore system configurations directly while the underlying operational problem may either remain solver-based or be incorporated into the evolutionary search.

The first two approaches can be separated along two dimensions: the scalarization formulation and the strategy used to select its parameters. Table 1 maps the reviewed literature according to these dimensions. Direct or hybrid evolutionary optimisation is shown separately, since it changes the variables explored by the multi-objective algorithm rather than only the parameters of a scalarised problem.

Weighted-sum optimisation has the broadest coverage. Besides conventional weight sweeps [18], more advanced methods have been used to select the weights. Tangi and Amaranto [19], for example, applied an outer evolutionary algorithm to search the

Table 1 Literature coverage of the selected multi-objective optimisation approaches.

Formulation	Systematic / predefined	Adaptive	Evolutionary
Weighted Sum	Weight sweep Established [18, 19]	Adaptive Weighted Sum Established [20]	Evolutionary weight search Calliope precedent [19]
Augmented Chebyshev	Weight sweep	Adaptive weight refinement	Evolutionary weight search Mainly MOEA literature
ε -Constraint / AUGMECON2	ε grid Established [21, 22]	Adaptive / refined search AUGMECON2 [11]	Evolutionary ε search No direct precedent identified
Direct / hybrid evolutionary system-design optimisation Established [19, 23]			

Direct / established precedent

Related or partial precedent

No directly relevant study identified

weight space of Calliope’s existing weighted-sum objective. This produced a better distributed Pareto approximation than predefined sampling. A related study further showed that evolutionary weight selection can achieve better front quality with fewer evaluations [24]. However, these approaches improve only the exploration of the weight space. The underlying formulation remains a weighted sum and therefore retains its inability to recover unsupported Pareto-optimal solutions from a non-convex attainable objective set.

The ε -constraint family approaches this problem from the formulation side. Finke and Bertsch [21] implemented AUGMECON [9] in Backbone while preserving the existing LP/MILP solver infrastructure, demonstrating its applicability even to energy-system models with several million variables and equations. The conventional ε -constraint method has also been applied in a Calliope-based district energy model [22]. In addition, AUGMECON2 reduces redundant solver runs by using information from previously obtained solutions to bypass unnecessary points of the ε grid [11]. Thus, both systematic and more refined exploration of the ε -constraint formulation are established, although no directly relevant evolutionary search over its parameters was identified.

Chebyshev scalarization is less extensively covered in the reviewed energy-system literature. Weighted and augmented Chebyshev methods, including adaptive parameter-selection procedures, are established in general multi-objective optimisation [12, 13]. However, their application to solver-based energy-system models is less common, and no corresponding implementation in Calliope was identified. This is particularly relevant because Chebyshev scalarization can recover Pareto-optimal solutions that weighted-sum optimisation cannot reach.

A completely separate approach is to apply evolutionary search directly to energy-system decisions. Prina et al. [23] used an evolutionary algorithm to explore investment decisions while retaining a solver-based operational model. Tangi and Amaranto [19] investigated a related approach with Calliope. Such methods avoid relying exclusively on scalarization parameters, but also change the optimisation architecture: the mathematical solver becomes part of the evaluation of candidate configurations generated by the outer algorithm.

Table 1 therefore shows that most parts of the considered design space have already been investigated individually. This does not, however, determine which approach performs better. Results from separate studies are difficult to compare because the underlying models, objectives, solvers, computational budgets, and evaluation criteria differ. Table 2 therefore considers a second question: which approaches have actually been compared within the same study?

Table 2 Cross-method comparisons identified in the reviewed multi-objective energy-system literature. Coloured curves connect approaches evaluated within the same study.

Formulation	Systematic / predefined	Adaptive	Evolutionary
Weighted Sum	Weight sweep Established [18, 19]	Adaptive Weighted Sum Established [20]	Evolutionary weight search Calliope precedent [19]
Augmented Chebyshev	Weight sweep	Adaptive weight refinement	Evolutionary weight search Mainly MOEA literature
ε -Constraint / AUGMECON	ε grid Established [21, 22]	Adaptive / refined search AUGMECON2 [11]	Evolutionary ε search No direct precedent identified
Direct / hybrid evolutionary system-design optimisation			
Established [19, 23]			

Tangi and Amaranto [19]	Areola et al. [25]	Hennen et al. [26]
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The most relevant comparison for Calliope is provided by Tangi and Amaranto [19]. They compared predefined weighted-sum sampling, evolutionary search over the same weights, and evolutionary search over system configurations. Evolutionary weight selection produced the most homogeneous and complete Pareto approximation in their case study, while direct evolutionary optimisation produced a less complete front despite a larger evaluation budget. This suggests that evolutionary algorithms can be useful not only as replacements for mathematical optimisation, but also as tools for guiding an existing scalarization.

Comparisons involving other formulations have also been performed outside Calliope. Areola et al. [25] compared weighted-sum and ε -constraint approaches with evolutionary algorithms, while Hennen et al. [26] compared conventional scalarizations with an adaptive Normal Constraint method. These studies demonstrate that both cross-method benchmarking and adaptive Pareto-front generation have already been considered in energy-system optimisation. However, as illustrated in Table 2, the comparisons cover only selected parts of the design space.

The resulting research gap is therefore more broad than a general lack of multi-objective methods. Weighted-sum optimisation, ε -constraint methods, Chebyshev-based methods, adaptive parameter selection, and evolutionary optimisation have all been studied to different degrees. What has not been identified in the reviewed literature is a controlled comparison that separates the effect of the scalarization formulation from the effect of the strategy used to explore its parameter space. In particular, weighted-sum, ε -constraint/AUGMECON, and Chebyshev formulations have not been compared within the same Calliope model under common solver settings, computational budgets, and evaluation criteria.

This distinction is important because performance differences may originate from either component. A method may produce a better Pareto front because its scalarization can reach additional solutions, because its parameters are explored more efficiently, or because it receives a larger computational budget. The proposed work therefore treats scalarization formulation and parameter-space exploration as separate experimental dimensions. Calliope 0.7.0 provides a suitable basis for this comparison, as its mathematical formulation can be extended through custom math definitions while retaining the existing optimisation backend [7].

6 Methodology

To-Do

to be discussed with Andreas/Dimeth

- Andreas
 - How did you analyse the literature to find promising approaches?
 - How did you identify promising approaches?
 - Select the most promising ones for theoretical comparison!
 - Compare them theoretically?
 - Set up hypotheses/expectation based on the theoretical comparison!
 - Implement, test and benchmark them!
 - Introduce the case study!
 - Run the implementation against the case study!
 - Analyse the results of the benchmark and case study in depth!

References

- [1] Pfenninger, S., Pickering, B.: Calliope: a multi-scale energy systems modelling framework. *Journal of Open Source Software* **3**(29), 825 (2018)
- [2] Sahoo, S., Joshi, M.Y., Ramirez Camargo, L.: A Fast-Processing Open-Source Building Energy Model for Energy Communities: 17th International Conference on Applied Energy, ICAE 2025. *Energy Proceedings* **65** (2026) <https://doi.org/10.46855/energy-proceedings-12247>
- [3] Sahoo, S., Ravani, J., Valdes, J., Camargo, L.R.: Integration of a building energy model with a regional planning tool for energy communities analysis. Technical Report EGU26-5363, Copernicus Meetings (March 2026). <https://doi.org/10.5194/egusphere-egu26-5363>. Conference Name: EGU26. <https://meetingorganizer.copernicus.org/EGU26/EGU26-5363.html>
- [4] Dantzig, G.B.: *Linear Programming and Extensions*. Princeton University Press, Princeton, NJ (1963). <https://doi.org/10.1515/9781400884179>
- [5] Nemhauser, G.L., Wolsey, L.A.: *Integer and Combinatorial Optimization*. John Wiley & Sons, New York (1988). <https://doi.org/10.1002/9781118627372>
- [6] Papadimitriou, C.H.: On the complexity of integer programming. *J. ACM* **28**(4), 765–768 (1981) <https://doi.org/10.1145/322276.322287>
- [7] Calliope contributors: Calliope 0.7.0: Base Mathematical Formulation. <https://calliope.readthedocs.io/en/v0.7.0.dev3/math/base/>. Accessed: 2026-08-02 (2024)
- [8] Hart, W.E., Watson, J.-P., Woodruff, D.L.: Pyomo: Modeling and solving mathematical programs in python. *Mathematical Programming Computation* **3**(3), 219–260 (2011) <https://doi.org/10.1007/s12532-011-0026-8>
- [9] Mavrotas, G.: Effective implementation of the ϵ -constraint method in Multi-Objective Mathematical Programming problems. *Applied Mathematics and Computation* **213**(2), 455–465 (2009) <https://doi.org/10.1016/j.amc.2009.03.037>
- [10] Das, I., Dennis, J.E.: A closer look at drawbacks of minimizing weighted sums of objectives for Pareto set generation in multicriteria optimization problems. *Structural optimization* **14**(1), 63–69 (1997) <https://doi.org/10.1007/BF01197559>
- [11] Mavrotas, G., Florios, K.: An improved version of the augmented ϵ -constraint method (AUGMECON2) for finding the exact pareto set in multi-objective integer programming problems. *Applied Mathematics and Computation* **219**(18), 9652–9669 (2013) <https://doi.org/10.1016/j.amc.2013.03.002>. Accessed 2026-08-03
- [12] Steuer, R.E., Choo, E.-U.: An interactive weighted Tchebycheff procedure for multiple objective programming. *Mathematical Programming* **26**(3), 326–344 (1983) <https://doi.org/10.1007/BF02591870>
- [13] Dächert, K., Gorski, J., Klamroth, K.: An augmented weighted tchebycheff method with adaptively chosen parameters for discrete bicriteria optimization problems. *Computers & Operations Research* **39**(12), 2929–2943 (2012) <https://doi.org/10.1016/j.cor.2012.02.021>
- [14] Fonseca, C.M., Fleming, P.J.: An overview of evolutionary algorithms in multiobjective optimization. *Evolutionary Computation* **3**(1), 1–16 (1995) <https://doi.org/10.1016/j.evco.1995.01.001>

[//doi.org/10.1162/evco.1995.3.1.1](https://doi.org/10.1162/evco.1995.3.1.1)

- [15] Zitzler, E., Thiele, L.: Multiobjective evolutionary algorithms: A comparative case study and the strength pareto approach. *IEEE Transactions on Evolutionary Computation* **3**(4), 257–271 (1999) <https://doi.org/10.1109/4235.797969>
- [16] Srinivas, N., Deb, K.: Multiobjective optimization using nondominated sorting in genetic algorithms. *Evolutionary Computation* **2**(3), 221–248 (1994) <https://doi.org/10.1162/evco.1994.2.3.221>
- [17] Deb, K., Pratap, A., Agarwal, S., Meyarivan, T.: A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation* **6**(2), 182–197 (2002) <https://doi.org/10.1109/4235.996017>
- [18] Eloranta, V., Inkeri, E., Luoranen, M., Grönman, A.: Balancing trade-offs in nordic smart local energy systems: Multi-objective and multi-criteria optimization for cost, emissions, and self-sufficiency. *Renewable Energy* **257**, 124696 (2026) <https://doi.org/10.1016/j.renene.2025.124696>
- [19] Tangi, M., Amaranto, A.: Advancements in multi-objective optimization for planning and management of multi-energy systems. *Journal of Sustainable Development of Energy, Water and Environment Systems* **13**(2), 1130584 (2025) <https://doi.org/10.13044/j.sdewes.d13.0584>
- [20] Kim, I.Y., Weck, O.L.: Adaptive weighted-sum method for bi-objective optimization: Pareto front generation. *Structural and Multidisciplinary Optimization* **29**(2), 149–158 (2005) <https://doi.org/10.1007/s00158-004-0465-1>
- [21] Finke, J., Bertsch, V.: Implementing a highly adaptable method for the multi-objective optimisation of energy systems. *Applied Energy* **332**, 120521 (2023) <https://doi.org/10.1016/j.apenergy.2022.120521>
- [22] Schilt, U., Vijayananda, S., Schneeberger, S., Meyer, M., Iyyakkunnel, S., Vecsei, P.M., Schuetz, P.: How can a geothermal storage system be optimally integrated into a local district? a case study. *Journal of Physics: Conference Series* **3140**(6), 062012 (2025) <https://doi.org/10.1088/1742-6596/3140/6/062012>
- [23] Prina, M.G., Casalicchio, V., Kaldemeyer, C., Manzolini, G., Moser, D., Wanitschke, A., Sparber, W.: Multi-objective investment optimization for energy system models in high temporal and spatial resolution. *Applied Energy* **264**, 114728 (2020) <https://doi.org/10.1016/j.apenergy.2020.114728>
- [24] Tangi, M., Amaranto, A.: Designing integrated and resilient multi-energy systems via multi-objective optimization and scenario analysis. *Applied Energy* **382**, 125281 (2025) <https://doi.org/10.1016/j.apenergy.2025.125281> . Accessed 2026-07-14
- [25] Areola, R.I., Adebiyi, A.A., Reddy, D.J.: Comparative benchmarking of multi-objective algorithms for renewable energy system design using pareto front quality metrics. *Applied Sciences* **16**(8), 3775 (2026) <https://doi.org/10.3390/app16083775>
- [26] Hennen, M., Voll, P., Bardow, A.: An adaptive normal constraint method for bi-objective optimal synthesis of energy systems. In: *Computer Aided Chemical Engineering* vol. 33, pp. 1279–1284 (2014). <https://doi.org/10.1016/B978-0-444-63455-9.50048-9>